

Exercise

Sunday, 16 August 2026 3:01 pm

3. Find the number(s) c referred to in the intermediate value theorem for each function over the interval indicated and for the given value of L .

(a) $f(x) = -x^2 + 2x + 3$ over $[-1, 0]$ using $L = 2$

Theorem 1.2 (Intermediate Value Theorem). Assume that $f \in C[a, b]$ and L is any number between $f(a)$ and $f(b)$. Then there exists a number c , with $c \in (a, b)$, such that $f(c) = L$.

(b) $f(x) = \sqrt{x^2 - 5x - 2}$ over $[6, 8]$ using $L = 3$

4. Find the upper and lower bounds referred to in the extreme value theorem for each function over the interval indicated.

(a) $f(x) = x^2 - 3x + 1$ over $[-1, 2]$

Theorem 1.3 (Extreme Value Theorem for a Continuous Function). Assume that $f \in C[a, b]$. Then there exists a lower bound M_1 , an upper bound M_2 , and two numbers $x_1, x_2 \in [a, b]$ such that

$$(7) \quad M_1 = f(x_1) \leq f(x) \leq f(x_2) = M_2 \quad \text{whenever } x \in [a, b].$$

We sometimes express this by writing

$$(8) \quad M_1 = f(x_1) = \min_{a \leq x \leq b} \{f(x)\} \quad \text{and} \quad M_2 = f(x_2) = \max_{a \leq x \leq b} \{f(x)\}.$$

5. Find the number(s) c referred to in Rolle's theorem for each function over the interval indicated.

Theorem 1.5 (Rolle's Theorem). Assume that $f \in C[a, b]$ and that $f'(x)$ exists for all $x \in (a, b)$. If $f(a) = f(b) = 0$, then there exists a number c , with $c \in (a, b)$, such that $f'(c) = 0$.

(a) $f(x) = x^4 - 4x^2$ over $[-2, 2]$

6. Find the number(s) c referred to in the mean value theorem for each function over the interval indicated.

Theorem 1.6 (Mean Value Theorem). Assume that $f \in C[a, b]$ and that $f'(x)$ exists for all $x \in (a, b)$. Then there exists a number c , with $c \in (a, b)$, such that

(11)
$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

(a) $f(x) = \sqrt{x}$ over $[0, 4]$

(b) $f(x) = \frac{x^2}{x+1}$ over $[0, 1]$

10. Find the number(s) c referred to in the mean value theorem for integrals for each function, over the interval indicated.

(a) $f(x) = 6x^2$ over $[-3, 4]$

(b) $f(x) = x \cos(x)$ over $[0, 3\pi/2]$

8. Apply the first fundamental theorem of calculus to each function over the interval indicated.

Theorem 1.8 (First Fundamental Theorem). If f is continuous over $[a, b]$ and F is any antiderivative of f on $[a, b]$, then

$$(12) \quad \int_a^b f(x) dx = F(b) - F(a) \quad \text{where } F'(x) = f(x).$$

(a) $f(x) = xe^x$ over $[0, 2]$

(b) $f(x) = \frac{3x}{x^2 + 1}$ over $[-1, 1]$

9. Apply the second fundamental theorem of calculus to each function.

(a) $\frac{d}{dx} \int_0^x t^2 \cos(t) dt$

(b) $\frac{d}{dx} \int_1^{x^3} e^{t^2} dt$

Theorem 1.9 (Second Fundamental Theorem). If f is continuous over $[a, b]$ and $x \in (a, b)$, then

$$(13) \quad \frac{d}{dx} \int_a^x f(t) dt = f(x).$$

Theorem 1.10 (Mean Value Theorem for Integrals). Assume that $f \in C[a, b]$. Then there exists a number c , with $c \in (a, b)$, such that

$$\frac{1}{b-a} \int_a^b f(x) dx = f(c).$$

The value $f(c)$ is the average value of f over the interval $[a, b]$.

Find the sum of each sequence or series.

(b) $\left\{ \frac{2}{3^n} \right\}_{n=1}^{\infty}$

(d) $\sum_{k=1}^{\infty} \frac{1}{4k^2 - 1}$

Find the Taylor polynomial of degree $n = 4$ for each function expanded about the given value of x_0 .

(a) $f(x) = \sqrt{x}$, $x_0 = 1$

Use synthetic division (Horner's method) to find $P(c)$.

(a) $P(x) = x^4 + x^3 - 13x^2 - x - 12, c = 3$